

Full length article

Re-examining the no-choice option in conjoint analysis

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ARTICLE INFO

Keywords:

Economic choice

Outside goods

Prices

Stable preferences

ABSTRACT

The validity of using conjoint analysis to conduct an economic evaluation of product characteristics rests on the inclusion of brand names, prices, and an outside “no-choice” option in the choice task. The no-choice option is included in case respondents determine that some other offering, not included in the conjoint choice task, is preferred to those that are included and that it would be better to hold onto their money and not make a purchase at that time. Selecting the no-choice option assumes that respondents have some level of knowledge of the value and prices of goods in the market. In this paper, we show that survey respondents may lack this information and make inferences about market prices from the conjoint exercise itself. This learning effect is especially problematic for new products for which a set of reference prices do not yet exist, but can also be problematic in established markets that are familiar. We discuss results from two sets of conjoint experiments, one in a new product category conducted in three countries in Europe, and another in an established category in the United States involving three experimental conditions that inform respondents about products and prices available in the marketplace. We find that the lack of knowledge of competitive offerings and prices affects estimates of brand values but not the value of other product features. In addition, we discuss aspects of how a well-designed conjoint study mitigates the effects of this type of learning in conjoint analysis.

1. Introduction

Conjoint analysis in marketing often takes an economic view of choice in which respondents can recall and construct their preferences for hypothetical offerings (Manski et al., 1981; Ben-Akiva et al., 2019). The minimal requirements for conducting a conjoint study to study marketplace demand is the inclusion of product brand and price, and an outside “no-choice” option. Brand is needed for respondents to imagine the offering and recall associated attributes and levels of performance. The brand name serves as a proxy for the unmentioned attributes of a brand in a conjoint study, and consumer knowledge of these features is what gives the brand names its value. When brand is absent, it is not possible to measure the competitive value of product features and economic analysis implicitly assumes offerings operate as monopolists, leading to an over-statement of value (Orme et al., 2023).

Prices provide a metric to reconcile utility estimates to a common scale and are needed to compute measures of economic value, such as willingness to pay. A conjoint analysis with only brand and price is often referred to as a brand-price tradeoff study (Rao, 2014). However, conjoint studies usually include other product attributes or features of interest to the analyst. Attributes that are

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Received 1 November 2024; Received in revised form 4 September 2025; Accepted 20 September 2025

Available online 18 October 2025

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familiar to and understood by respondents factor into, or influence, their choices, but those that are unfamiliar or not understood are less important in the decision process (Balbontin et al., 2017; Sandorf et al., 2017).

The no-choice option in a conjoint study allows respondents to choose something other than one of the brands included in the choice task (Brazell et al., 2006). Not only does the no-choice option serve to increase the realism of the respondent's decision, but it can also improve the resulting market share and volume predictions from the analysis (Carson et al., 1994) and eliminate statistical biases (Haaijer et al., 2001). By selecting the no-choice option, respondents indicate a preference to opt out of the choice task and spend their money elsewhere. This allows respondents to compare the utility of the alternatives in each choice task to some fixed level of utility they expect to achieve in the market (Louviere et al., 2010; Bahamonde-Birke et al., 2017). Early attempts to model the no-choice option associated it with its own error component to relax substitution patterns among the choice alternatives (Scarpa et al., 2005).

We examine the role of this no-choice option in a marketing context to assess the determinants of demand in a competitive market, where all goods sold have a brand name and a price as a product attribute. The outside good typically represents the decision to not choose any of the choice alternatives in the choice task in the analysis, whereas in other domains such as health care (Salkeld et al., 2000; Becker and Zweifel, 2008) or transportation (Rose et al., 2008; Train and Wilson, 2008; Krueger et al., 2016), it generally reflects the status quo (e.g., continuing current treatment or sticking with an existing travel mode). Researchers are encouraged to include an outside good in conjoint studies to avoid forcing respondents to choose when they have the opportunity to defer choices or maintain their current situation in their daily lives (Boxall et al., 2009; Van Loo et al., 2015).

Various psychological factors that claim to violate economic assumptions have been shown to be associated with the decision to select the no-choice option in behavioral studies (Tversky and Shafir, 1992; Dhar, 1997; Gunasti and Ross, 2008), including the types of alternatives that are more likely to lose choice share to no-choice options (Dhar and Simonson, 2003), and how the inclusion of a no-choice option changes the decision process (Parker and Schrieff, 2011). Although the existing literature has focused on factors related to general choice deferral, we focus on the no-choice option in conjoint analysis. Consider, for example, a conjoint study of the determinants of elderly persons selecting an assisted living care facility. It is doubtful that many people are aware of different care options, nor of the daily rate of assisted care, unless they are involved in the financing or care of an elderly person. Respondents probably have a much better grasp of the incremental value of increased services, such as better dining options. Since the selection of the no-choice option depends on the overall price levels of the inside good options, there is a chance that respondents make inferences about market prices from the choice tasks themselves when they are unfamiliar with the product category.

The effect of choice task uncertainty has been shown to produce contextual effects documented in the behavioral economics literature, such as attraction and compromise effects (Natenzon, 2019; Khaw et al., 2020; Enke and Graeber, 2023; He, 2024). Sometimes the uncertainty is due to the complexity of the choice task (Boxall et al., 2009) and sometimes it is due to imprecisely defined attribute-levels, such as stating that the performance of product along a specific dimension is "low", without providing a precise definition of what this means. Conjoint studies attempt to mitigate the effects of uncertainty by screening out respondents who are not familiar with the product category and by providing concise and concrete definitions of product features (Orme, 2010). Screening questions provide some degree of assurance that consumers are likely aware of product features and general price levels because of their past or intended purchase activities. However, there are instances when this is difficult to accomplish, such as when the entire product category is new to the market. In this setting, the "stable preference" assumption of conjoint analysis may not be true and respondents may learn about product features and prices from the conjoint exercise itself.

The purpose of this paper is to examine respondent learning about marketplace prices and product features in a conjoint exercise where the value of an offering is based on the concept of a random utility model (Louviere et al., 2010). If standard screening questions about product participation are sufficient for identifying qualified candidates who are aware of marketplace prices and product features, then providing additional pricing and attribute information will minimally affect part-worth estimates. However, if survey screening questions are an insufficient proxy for brand price and attribute knowledge, then providing price and attribute information may change consumer preferences for the outside good and possibly other product features. Invariance of consumer choices to outside good information is therefore fundamentally related to the assumption that the economic view of preference construction is valid for conjoint studies with properly screened respondents.

The organization of this paper is as follows. Section 2 provides a review of assumptions used in choice-based conjoint analysis. Section 3 describes a conjoint study in a new product category and a second study in an established product category in which respondents are exposed to alternative sets of information about market offerings and prices. Section 4 provides model-free evidence of consumers updating preferences for the outside good in the first study and develops a model for the outside good. Empirical results for the two studies are reported in Section 5, Section 6 provides a discussion of our results and concluding remarks are offered in Section 7.

2. Basic assumptions

The discrete choice model typically used in conjoint analysis is characterized by one and only one choice alternative being selected at a time. Consumers are assumed to be endowed with a budget for the purchase that determines the upper limit of expenditure they are willing to make. If the good costs less than the budgeted amount, which we denote " E " for expenditure, then the remainder of the unspent money ($E - p$) can be used for other purposes. Thus, we conceive of choice as between a number of J different "inside" goods and an "outside" good that represents money unspent in the choice task. The utility function for this situation can be represented as:

$$u(x, z) = \sum_{k=1}^J \psi_k x_k + \psi_z z \quad (1)$$

where x is a vector of demand for the inside goods, z is the demand for the outside good or non-purchase, ψ_k is the marginal utility of choosing the k -th good and ψ_z is the marginal utility for money. It is customary in choice models to include an error term to the above specification to allow for unobserved factors affecting choice. The utility from selecting each of the alternatives as:

$$\begin{aligned} u(x_1 = 1, z = E - p_1) &= \psi_1 + \psi_z(E - p_1) + \varepsilon_1 \\ u(x_2 = 1, z = E - p_2) &= \psi_2 + \psi_z(E - p_2) + \varepsilon_2 \\ &\vdots \\ u(x_J = 1, z = E - p_J) &= \psi_J + \psi_z(E - p_J) + \varepsilon_J \\ u(x = 0, z = E) &= \psi_z(E) + \varepsilon_z \end{aligned}$$

The utility for each of the inside goods is comprised of three terms – (i) the marginal utility for the inside good; (ii) utility for the unspent money that can be put to other use; and (iii) an error term. The utility for the outside good is just the utility for the budgeted allotment, E , plus error. Consumers are assumed to select the choice alternative that provides them greatest utility. For ease of illustration, let the outside good be indexed as the 0-th inside good, with $\psi_0 = 0$ and $p_0 = 0$. Let $V_k = \psi_k + \psi_z(E - p_k)$ denote the deterministic part of the k th inside good, for $k = 0, \dots, J$. Then the choice model becomes:

$$\begin{aligned} \Pr(j) &= \Pr(\psi_j + \psi_z(E - p_j) + \varepsilon_j > \psi_k + \psi_z(E - p_k) + \varepsilon_k \text{ for any } k \in \{0, \dots, J\}/\{j\}) \\ &= \Pr(V_j + \varepsilon_j > V_k + \varepsilon_k \text{ for any } k \in \{0, \dots, J\}/\{j\}) \\ &= \Pr(\varepsilon_k < V_j - V_k + \varepsilon_j \text{ for any } k \in \{0, \dots, J\}/\{j\}) \\ &= \int_{-\infty}^{+\infty} \left[\int_{-\infty}^{V_j - V_0 + \varepsilon_j} \dots \int_{-\infty}^{V_j - V_J + \varepsilon_j} \phi(\varepsilon_J) \dots \phi(\varepsilon_0) \right] \phi(\varepsilon_j) d\varepsilon_J \dots d\varepsilon_0 d\varepsilon_j \\ &= \int_{-\infty}^{+\infty} \prod_{k=0, k \neq j}^J \Phi(V_j - V_k + \varepsilon_j) \phi(\varepsilon_j) d\varepsilon_j \end{aligned}$$

where Φ denotes the cdf and ϕ denotes the pdf of the distribution of ε . Distributional assumptions play a role in determining the functional form of the choice model, with extreme value errors leading to the logit model and normally distributed errors giving rise to the probit model. Assuming standard extreme value errors (i.e., $EV(0, 1)$) results in the following logit expression for the choice probability.

$$\begin{aligned} \Pr(j) &= \frac{\exp[V_j]}{\exp[V_0] + \sum_{k=1}^J \exp[V_k]} \\ &= \frac{\exp[\psi_j + \psi_z(E - p_j)]}{\exp[\psi_z(E)] + \sum_{k=1}^J \exp[\psi_k + \psi_z(E - p_k)]} \\ &= \frac{\exp[\psi_j - \psi_z p_j]}{1 + \sum_{k=1}^J \exp[\psi_k - \psi_z p_k]} \end{aligned} \quad (2)$$

Thus, the choice probability is a function of a choice-specific intercept (ψ_k) and a price term with a coefficient that is common across the choice alternatives. The deterministic portion of the outside good is assigned a value of zero, resulting in the “+1” in the denominator. The budget allotment E cancels out of the expression and is often not considered in conjoint studies (cf. [Pachali et al. \(2017\)](#), [Kim et al. \(2022\)](#), [Hung et al. \(2025\)](#)). Conjoint analysis for discrete choice further assumes that the marginal utility for a good, ψ_j , is a linear function of brand attributes and their levels:

$$\psi_j = \beta' a_j \quad (3)$$

where $a_j = [a_{j1}, \dots, a_{jL}]'$ is the vector of attribute levels for good j , with L possible attribute levels excluding price. The attributes are often coded using a dummy variable specification where one of the attribute-levels is selected as the null level of the attribute, or using effects-coding that constrain the sum of coefficients to equal zero for an attribute. For either specification, conjoint analysis measures the value of changes among the levels of an attribute. Since the valuation of the attribute-levels are jointly determined through Eq. (3), the marginal utilities for a respondent are comparable across the attributes and features included in the analysis. The marginal utility of a product feature (i.e., its part-worth) can therefore be compared to the utility of changes in the levels of other attributes, including price (see also [Green and Rao, 1971](#)).

Respondent heterogeneity can be incorporated into conjoint analysis using a variety of random-effect models. The simplest and most widely used is a Normal model for heterogeneity. Denoting all individual-level part-worth parameters β_h and marginal utility of money ψ_{zh} for respondent h we have:

$$(\beta_h, \psi_{zh})' \sim \text{Normal}(\bar{\beta}, \Sigma) \quad (4)$$

We obtain parameter estimates using hierarchical Bayesian Markov chain Monte Carlo methods. Alternatively, one could use classical methods of estimation ([Train, 2009](#)) using error components ([Scarpa et al., 2005](#); [Fiebig et al., 2010](#)). In our analysis reported below, diffuse but proper prior distributions were specified for all model parameters.

Table 1
MP3 player study sample size and market penetration.

	France	Italy	UK
Number of respondents	482	452	487
Market penetration	3.4%	2.2%	5.3%

Table 2
MP3 player products attributes and levels.

Attribute	Levels						
	1	2	3	4	5	6	7
Brand	Samsung	Apple	Sony	Creative	–	–	–
Shape	(t10)	iPod nano	iPod Touch	p2	–	–	–
LCD display size	(2.0")	3.0"	3.3"	–	–	–	–
Touch Screen	(No)	Yes	–	–	–	–	–
Mobile TV	(No)	Yes	–	–	–	–	–
Satellite Navigation	(No)	Yes	–	–	–	–	–
Wireless Connection	(None)	Wi-Fi	Bluetooth	Wi-Fi/Bluetooth	–	–	–
Storage Capacity	(4 GB)	8 GB	16 GB	32 GB	–	–	–
Price (GBP)	179	199	249	300	399	499	599

Note: The attribute levels in the parentheses are used as the reference level of each corresponding attribute.

3. Data

We examine the effect of respondent knowledge of market prices on part-worth estimates using two conjoint studies and building on the standard conjoint model described above. The first study examines the stability of preferences for product features in a new market where respondents were unlikely to know the prices of offerings. We find that the value of the outside good substantially increases and that the brands lose value across the conjoint choice tasks. The second study is for offerings in a mature product category in which respondents are more likely to be aware of market prices. Here we find that presenting more information about prices and features has little impact on stabilizing the estimated preference for the outside good.

3.1. MP3 player study

We begin our investigation using data from a MP3 player study conducted in 2008 when this product was new to the market and market prices were unlikely to be known by respondents. The seven attributes were chosen based on the technical innovations planned by the client for the upcoming year, and the prices were determined according to these new features for 2009. The brands reflect the current competition in the three countries. At the time the study was conducted, one of the key innovations in the market was the ability to play videos on MP3 players. Additionally, it is important to note that the iPhone was not yet available in these markets—it had only been announced for Germany and the UK in the following months. Table 1 reports the number of respondents from each country and the market penetration of the MP3 players in each at the time the study was fielded.

Fifteen sets of choice tasks, each with 15 tasks, were generated using a SAS algorithm that produces D-efficient, orthogonal and level-balanced experimental designs (Kuhfeld, 2010). The 15 sets of choice tasks were randomly assigned to the respondents and the order of the 15 choice tasks within these 15 sets was also randomized, so there would be no influence on the outside good due to the design. Before the experiment began, all attributes and variations were thoroughly presented and explained to the respondents through text and visual materials. The study was conducted in France, Italy and the United Kingdom, providing us the ability to replicate results across countries. There are four choice options including the outside good in each choice task.

Table 2 reports the attributes and levels used in the study. There are four brands comprising the set of inside goods and a total of nine different attributes. Price is reported in units of GBP (Great British Pound), although tasks in each country were displayed in local currency; for the analysis, price is scaled in units of 100. The first column of attribute-levels is listed parenthetically to indicate reference levels for each attribute. Part-worths for the remaining attribute-levels represent the marginal utility of the indicated level relative to the reference level for each attribute. The conjoint design was randomized across respondents so that different respondents were exposed to the choice tasks in a different order.

3.2. Tooth whitening study

Participants were recruited from a United States panel for the tooth whitening study conducted in 2019. This category was selected for analysis because tooth whitening products were first introduced into the market in the 1970's and these products are widely used today.¹ Tooth whitening products also exhibit large price variation and there is a variety of product attributes. A list of product attributes used in the survey is presented in Table 3 together with their definitions.

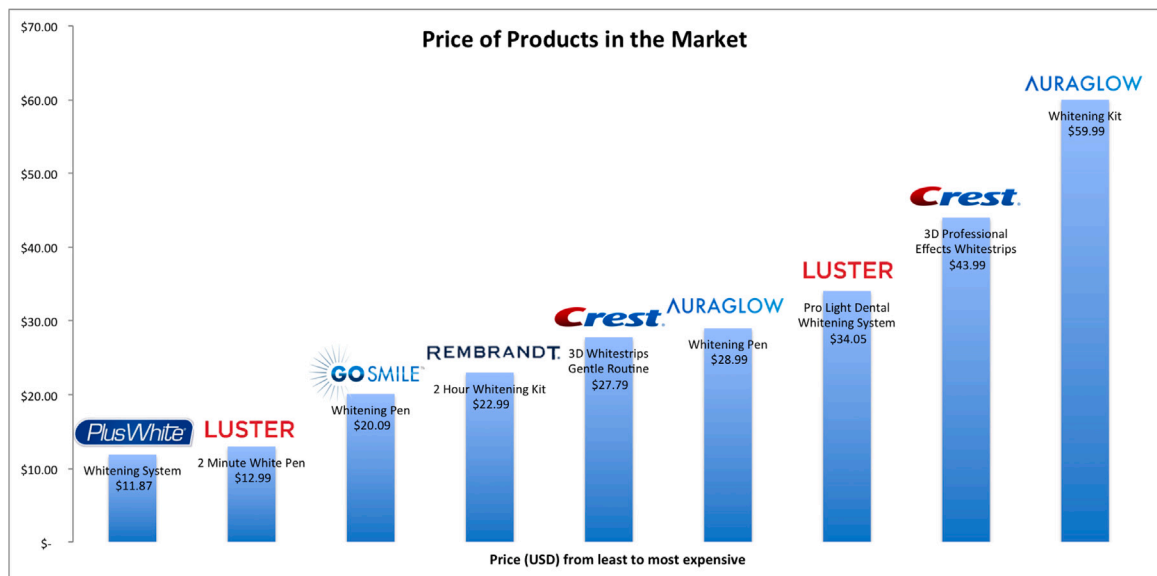
¹ <https://www.harrisonburgsmiles.com/blog/2022/04/06/history-and-evolution-of-teeth-whitening/>

Table 3

Tooth-whitening study product attributes, definitions, and levels.

Attribute	Definition	Levels							
		1	2	3	4	5	6	7	8
Brand	The brand name of the teeth-whitening product	Crest	PlusWhite	Rembrandt	GoSmile	Auraglow	Luster	–	–
Form	The method of application of the product	(Strips)	Whitening Pen	Trays & Gel	Light Tech	–	–	–	–
Treatment Time	The total suggested time for a single whitening treatment	5	15	25	–	–	–	–	–
Number of Treatments	The total suggested number of whitening treatments to be completed	7	14	21	–	–	–	–	–
Time Until Results	The claimed amount of time until consumers should see visibly whiter teeth	3	7	14	–	–	–	–	–
Peroxide %	The percentage of the active ingredient (hydrogen peroxide) contained in the product	6	7	8	9	10	11	12	13
Price	The price of the product	14.99	24.99	34.99	44.99	54.99	–	–	–

Note: The attribute levels in the parentheses are used as the null level of each corresponding attribute.

**Fig. 1.** Price distribution graphic.

Individuals who responded to the invitation from the Internet panel provider to participate were presented with a series of screening questions to determine whether the potential respondent was “in” the product category and had sufficient knowledge to provide informative answers to the survey questions. To be included in the survey, respondents must make their own hygiene purchases, be medically qualified to use the product, and be engaged in or actively considering buying an offering in the product category. The screening questions used in the survey are presented in Appendix.

The qualified respondents who successfully passed the screening questions were required to watch a short video describing the choice task and defining the product attributes and their levels.² Respondents were then randomly assigned to one of the experimental conditions. The first condition did not provide any information about the prices or attribute of the marketplace and serves as a control group for the other two conditions.

The second experimental condition provided the respondents with the range of prices for the different brands under study, as well as brands not included in the survey.³ Prices were obtained from the prices posted on Amazon.com. Fig. 1 displays the information provided to respondents.

² see https://go.osu.edu/conjoint_instructions.

³ see https://go.osu.edu/second_experimental_condition

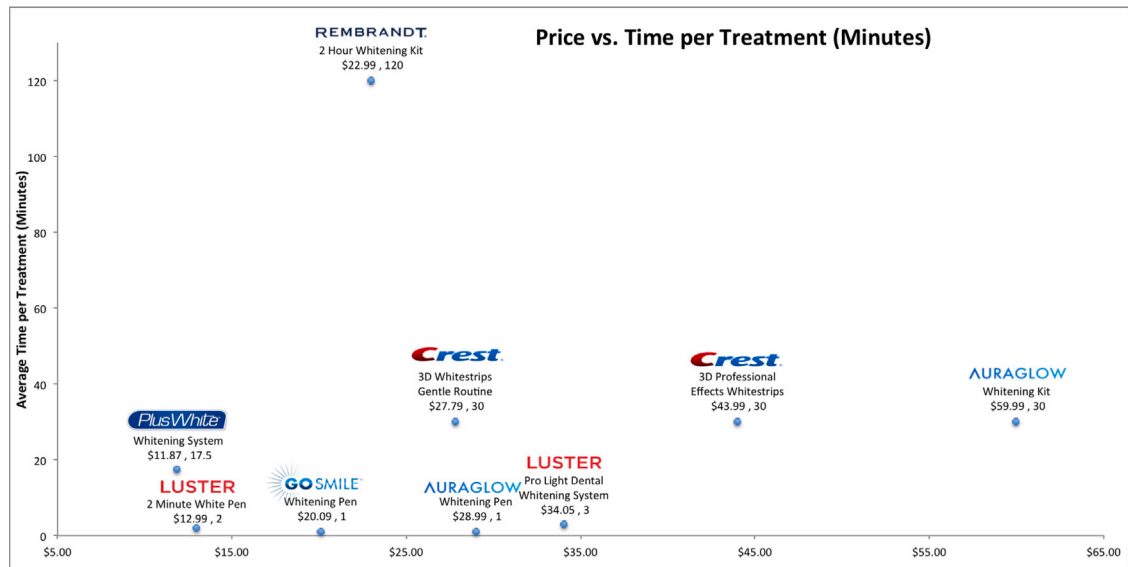


Fig. 2. Price vs. time per treatment graphic.

	Option A	Option B	Option C	I would not choose any of these.
Brand				
Form	Pen	Trays + Gel	Light Technology	
Time for One Treatment	5 minutes	15 minutes	15 minutes	
Number of Treatments	7	14	21	
Time to Results	3 days	7 days	3 days	
Percent Peroxide	12%	8%	10%	
Price	\$14.99	\$14.99	\$34.99	-

Fig. 3. Tooth whitening choice task.

The third experimental condition provided information about the relationship between brand, price, and effectiveness attributes.⁴ Graphs comparing price with time per treatment, number of treatments, time to results, and percentage of peroxide were explained to respondents and shown in succession. Fig. 2 displays the price versus time per treatment graphic shown to respondents.

Then, the conjoint choice tasks were presented to the qualified respondents, asking them to identify their most preferred offering among those chosen. Each choice task included three different brand names and a no-choice option from which respondents indicated their preferred brand. Each respondent was asked to indicate their preference in 12 choice tasks. An example choice task is shown in Fig. 3.

The survey ended with a series of demographic variables to help assess the representativeness of the sample population. A total of 1,141 qualified respondents completed the survey and provided information on their preferences. The number of respondents was evenly divided between the three experimental conditions.

⁴ see https://go.osu.edu/third_experimental_condition.

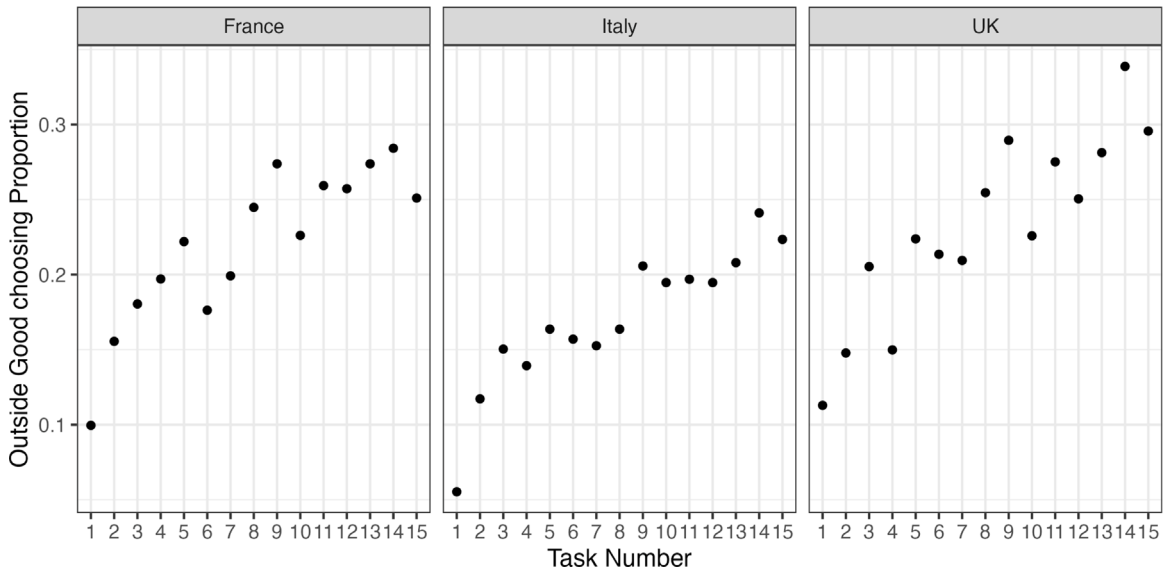


Fig. 4. Proportion of choosing the outside good option in each task from the MP3 player study.

4. Model

We begin our model development by examining patterns in responses to the conjoint tasks in the MP3 data set where we find respondents increasingly select the outside good over time. We relate this increase to the expected maximum utility of the choice set and to the task number and find that the respondents are dynamically adjusting their preferences. This information is used to guide our model development.

4.1. Preliminary analysis

Model-free evidence of respondents updating their preference for the outside good is provided in Fig. 4. The percentage of choices for the outside good is plotted across the choice tasks for each country in the MP3 study. A significant upward trend in the tendency of respondents to select the outside good is detected, indicating that the outside good does not exhibit constant utility across the choice tasks.

A simple model for the choice of the outside good is to assume that it is inversely related to the maximum attainable utility of the inside goods and a linear time trend for the task number. Conditioning on posterior draws of the conjoint part-worths, this results in a binomial logit model of the form:

$$\text{logit}(p_{ht}^{\text{out}}) = \alpha_0 + \alpha_1(W_{ht}) + \alpha_2(\text{task number}) \quad (5)$$

where p_{ht}^{out} is the probability of selecting the outside good for respondent h in choice task t , W_{ht} is the inclusive value of the inside good in the choice task and the task number takes on integer values from 1 to 15, the number of choice tasks. The inclusive value for the inside goods is equal to:

$$W_{ht} = \ln \sum_{k=1}^J \exp(\beta'_h a_{hkt} - \psi_{zh} p_{kt}) \quad (6)$$

where, in this analysis, (β_h, ψ_{zh}) is the posterior part-worth vector for each respondent h , estimated using the standard model specified by Eqs. (2), (3), and (4), a_{hkt} denotes the vector of attribute levels, and p_{kt} the price of the k -th inside good out of a total of J inside goods in choice task t .

Note that if the standard model is correct, the probability of choosing the outside good for respondent h in task t is

$$p_{ht}^{\text{out}} = \frac{1}{1 + \sum_{k=1}^J \exp(\beta'_h a_{hkt} - \psi_{zh} p_{kt})} = \frac{1}{1 + \exp(W_{ht})}$$

$$\Rightarrow \text{logit}(p_{ht}^{\text{out}}) = -W_{ht}$$

indicating that the selection of the outside good should be inversely related to the expected maximum utility of the inside goods in the choice task. We obtain estimates of the inclusive value W_{ht} for each respondent (h) and choice task (t), estimate the coefficients in Eq. (5) using respondent choices as the dependent variable, and average these coefficients over the Monte Carlo draws of the posterior distribution of part-worths in Eq. (4). Details are provided in Appendix.

Table 4
No-choice logistic regression coefficients (standard deviation).

	France	Italy	UK
Intercept (α_0)	-0.91 (0.06)	-1.06 (0.08)	-1.02 (0.06)
Inclusive Value (α_1)	-1.00 (0.03)	-1.03 (0.03)	-1.01 (0.03)
Time Trend (α_2)	0.11 (0.01)	0.13 (0.01)	0.12 (0.01)
Hit-Probability	0.821	0.845	0.817

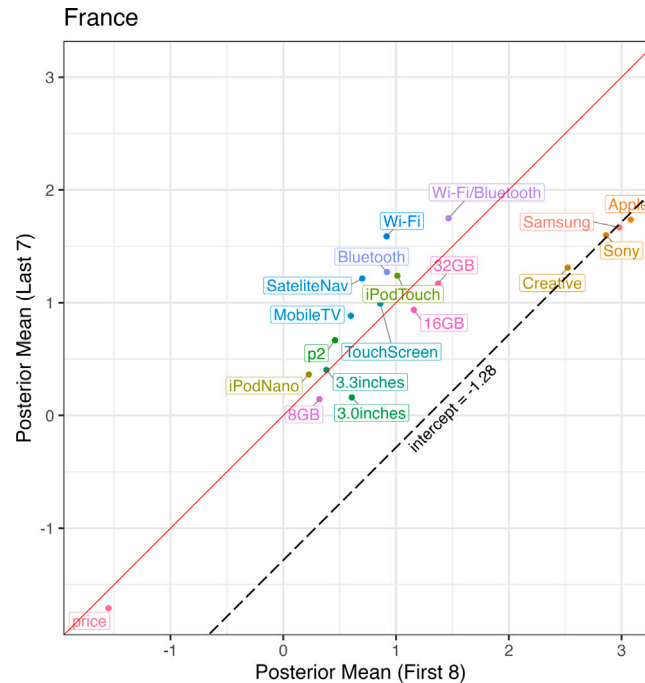


Fig. 5. Comparison of mean of random-effect distribution of part-worths from the first 8 choice tasks ($t = 1, \dots, 8$, x-axis) to last 7 tasks ($t = 9, \dots, 15$, y-axis) from France.

The results of our initial analysis, reported in Table 4, indicate that the probability of selecting the outside good is inversely related to the inclusive value of the inside goods, as expected, and also the task number, which was not expected. The results confirm that the selection of the outside good increases over time as respondents become more familiar with the conjoint exercise. That is, respondents appear to learn about the value of the offerings for money over time.

Due to the random task design assigned to each respondent, we can provide additional insight into the effect of respondent learning by dividing the dataset into two parts based on the task number. The first dataset consists of the first 8 tasks completed by each respondent ($t = 1, \dots, 8$), while the second dataset consists of the last 7 tasks ($t = 9, \dots, 15$). Fig. 5 compares conjoint part-worth estimates from these datasets with a 45-degree line that indicates equality. Aside from the part-worths for the brands (Samsung, Apple, Sony, Creative), the results indicate equality of the part-worth estimates. We also find similar results from Italy and the UK. The part-worths for the brands are not on the 45-degree line and instead are plotted on a second parallel line in the figure. This implies that the brand part-worths are estimated to be similar relative to each other, but at a different overall level. Since the outside good serves as the reference level for the brand intercepts, we conclude that the utility of the outside good is responsible for the upward trends shown in Fig. 4. Our results are also consistent with (Hess et al., 2012) who find that respondents often do not exhibit fatigue across stated preference tasks, which would affect all part-worths and not just respondent utility for the outside good.

Our preliminary analysis indicates that respondents' change their utility value for some (i.e., the brand coefficients) but not all (i.e., other product attributes and levels) of the conjoint model coefficients. In our proposed model, discussed below, we allow respondents to update only the outside good utility, while keeping the other attribute-level parameters constant. The result in Fig. 5 shows that keeping other attribute-level parameters constant is not a strong assumption and supports the sufficiency of our proposed model.

4.2. Proposed model

Our preliminary analysis of the MP3 data indicates that the selection of the no-choice option is affected by the quality of the inside goods and a time-varying effect (see Table 4). We combine these two effects in a model of dynamic updating.

Our proposed model introduces an additional term, $\psi_{out,t}$, to account for respondents dynamically updating their perception of the outside good utility throughout the tasks. Thus, the proposed specification becomes:

$$u_{out,t} = \psi_{out,t} + \psi_z(E) + \varepsilon_{out,t}, \quad \varepsilon_{out,t} \sim EV(0, 1).$$

where $u_{out,t}$ is the utility for the outside good in task t and $\psi_{out,1} = 0$ for identification. We propose that after making a choice in each task, the respondent observes the best inside good option and uses it to probabilistically update their beliefs about the utility of the outside good in the next time period. Since the analyst does not know for certain which option is best, our model assumes that updating occurs based on the maximum expected utility, $\max\{u_t\}$, of the choice set in task t , where u_t is the J -dimensional vector of utilities for the inside goods. For each task $t = 1, \dots, T$, the respondent observes the maximum utility of choosing any inside good, denoted as $\max\{u_t\}$, where

$$\begin{aligned} \max\{u_t\} &= \max_{k=1, \dots, J} \{u_{kt}\} \\ &= \ln \left\{ \sum_{k=1}^J \exp(\beta' a_{kt} + \psi_z(E - p_{kt})) \right\} + \varepsilon_{max,t} \\ &= \ln \left\{ \sum_{k=1}^J \exp(\beta' a_{kt} - \psi_z p_{kt}) \right\} + \psi_z(E) + \varepsilon_{max,t}, \quad \varepsilon_{max,t} \sim EV(0, 1), \end{aligned}$$

We further assume that the probability of updating is constant across the choice tasks, and respondents probabilistically substitutes the deterministic part of the outside good utility with:

$$\psi_{out,t+1} = \begin{cases} \ln \left\{ \sum_{k=1}^J \exp(\beta' a_{kt} - \psi_z p_{kt}) \right\}, & \text{with probability } \frac{\exp(\rho)}{1 + \exp(\rho)}, \\ \psi_{out,t}, & \text{otherwise.} \end{cases} \quad (7)$$

where ρ is a parameter that controls the updating probability after each choice task.

To determine whether our proposed model is necessary to describe the phenomenon across tasks, we evaluate an alternative model that allows the outside good utility to increase linearly through tasks:

$$\psi_{out,t} = e^{\gamma}(t-1), \quad t = 1, \dots, T, \quad (8)$$

where e^{γ} describes the magnitude of the increasing trend in the outside good utility across tasks. In this alternative linear model, respondents may experience outside good utilities that monotonically increase with the choice tasks. We constrain the slope to be non-negative because some respondents are observed to never choose the outside good and we want to avoid characterizing their behavior as being due to learning. These respondents are, in effect, constrained to follow the traditional model.

We incorporate respondent heterogeneity into our analysis using a hierarchical Bayes specification. Let β_h denote the conjoint part-worths, ψ_{zh} represent the price coefficient or marginal utility of money, and ρ_h be the parameter controlling the probability of substitution after each task for respondent h . For our proposed model, we have

$$\phi_h = (\beta'_h, \psi_{zh}, \rho_h)' \sim \text{Normal}(\bar{\phi}, \Sigma_{\phi}). \quad (9)$$

and for the alternative linear model we substitute γ_h into the model of heterogeneity:

$$\tilde{\phi}_h = (\beta'_h, \psi_{zh}, \gamma_h)' \sim \text{Normal}(\bar{\tilde{\phi}}, \tilde{\Sigma}_{\phi}). \quad (10)$$

Parameter estimates are obtained using Markov chain Monte Carlo methods, with all model parameters specified using diffuse but proper prior distributions. Further details are provided in Appendix.

5. Results

We compare the performance of our proposed model to the standard model and the model with the linear time trend in terms of model fit, parameter estimates, and inferences about the outside good.

5.1. Model fit

We measure the fit of the models with the log-marginal density (LMD) statistic using the Newton–Raftery (1994) approximation. Table 5 reports the fit for the standard conjoint choice model (Eq. (2)), the linear model for the outside good (Eq. (8)) and our proposed model (Section 4.2) in the two studies. The result from the MP3 Player and tooth whitening studies show that our proposed model, where the utility of the outside good is probabilistically updated, fits the data the best.

Table 5
Model fit
(Log Marginal Density).

Study	Model	Dataset		
MP3		France	Italy	UK
	Standard model	−3932	−3891	−3965
	Linear model	−3767	−3756	−3818
	Proposed model	−3678	−3643	−3663
Tooth Whitening		Condition 1	Condition 2	Condition 3
	Standard model	−1656	−1766	−1973
	Linear model	−1584	−1702	−1920
	Proposed model	−1470	−1634	−1824

Table 6
MP3 posterior mean (standard deviation) of random effects $\bar{\phi}$ for proposed model.

Attribute	France	Italy	UK
Brand			
Samsung	4.62 (0.62)	5.26 (0.52)	6.15 (0.77)
Apple	4.70 (0.62)	5.30 (0.51)	6.49 (0.76)
Sony	4.54 (0.62)	5.33 (0.53)	6.39 (0.77)
Creative	4.30 (0.62)	5.12 (0.53)	5.80 (0.76)
Shape			
iPodNano	0.30 (0.13)	0.13 (0.12)	−0.01 (0.12)
iPodTouch	1.07 (0.15)	0.62 (0.12)	0.62 (0.12)
p2	0.47 (0.12)	0.04 (0.10)	0.11 (0.11)
LCD display size			
3.0inches	0.44 (0.09)	0.23 (0.09)	0.46 (0.09)
3.3inches	0.38 (0.09)	0.26 (0.09)	0.43 (0.09)
Touch Screen	0.86 (0.10)	0.83 (0.09)	0.77 (0.09)
Mobile TV	0.75 (0.09)	0.62 (0.09)	0.95 (0.10)
Satellite Navigation	0.90 (0.10)	0.87 (0.10)	0.72 (0.09)
Wireless Connection			
Wi-Fi	1.23 (0.13)	1.07 (0.12)	1.21 (0.12)
Bluetooth	1.01 (0.12)	0.98 (0.11)	1.10 (0.12)
Wi-Fi/Bluetooth	1.50 (0.13)	1.39 (0.12)	1.47 (0.13)
Storage Capacity			
8 GB	0.31 (0.11)	0.33 (0.11)	0.44 (0.11)
16 GB	0.99 (0.12)	0.75 (0.11)	1.20 (0.12)
32 GB	1.24 (0.13)	1.13 (0.12)	1.59 (0.13)
Price	−1.56 (0.09)	−1.29 (0.07)	−1.71 (0.09)
ρ	−4.16 (0.23)	−4.30 (0.21)	−4.00 (0.21)

5.2. Parameter estimates

Tables 6 and 7 report the average part-worth estimates for the attribute levels, price, and the outside good updating parameter ρ of our proposed model. Estimates are reported in terms of the mean of the distribution of random effects $\bar{\phi}$. There is general agreement in the estimates between the data sets in each study. The MP3 study offers a setting to replicate the effect of the outside good on demand in a new product market since the effect occurs in each country. In addition, the tooth-whitening study provides a basis for inferences about different amounts of information about the outside good because results can be compared with data from the same market.

The primary difference in the estimates is in terms of the updating parameter ρ in Table 6 and the brand intercepts and price coefficients in Table 7. We find that the updating parameter is larger in the MP3 datasets than in the tooth-whitening datasets, indicating that respondents are updating their preference for the outside good to a greater extent as they progress through the survey. In addition, the brand intercepts in the tooth-whitening datasets are larger in Condition 1 than in Condition 3, indicating that the additional information provided in Condition 3 increases the likelihood of selecting the outside good. To illustrate the difference between the ρ estimates in the MP3 and tooth-whitening datasets, we calculate the update probability of at least once by the end of the survey, given by the function $f(\rho) = 1 - \left(1 - \frac{\exp(\rho)}{1 + \exp(\rho)}\right)^{\text{number of tasks} - 1}$. This evaluates to 0.195 for France in the MP3 dataset and 0.067 for Condition 1 in the tooth-whitening dataset by substituting $\rho = -4.16$ for the former and $\rho = -5.06$ for the latter.

The probability of updating at least once in the tooth-whitening dataset is similar across condition 1 ($Pr = 0.067$), condition 2 ($Pr = 0.078$), and condition 3 ($Pr = 0.071$). However, updating is a function of the both the frequency of updating and the amount of learning. Since respondents in Condition 3 are required to watch three glossary videos, they are likely better informed about market prices at the beginning of the conjoint exercise and may exhibit less learning, despite having similar updating probabilities.

Table 7Tooth-whitening posterior mean (standard deviation) of random effects $\bar{\phi}$ for proposed model.

Attribute	Condition 1	Condition 2	Condition 3
Brand			
Crest	34.13 (1.24)	32.34 (1.74)	28.30 (1.42)
PlusWhite	31.38 (1.12)	29.26 (1.61)	26.31 (1.37)
Rembrandt	33.39 (1.16)	30.71 (1.62)	28.02 (1.39)
GoSmile	32.60 (1.16)	31.33 (1.72)	27.83 (1.49)
Auraglow	32.06 (1.25)	30.78 (1.68)	27.80 (1.45)
Luster	31.51 (1.17)	29.62 (1.66)	26.82 (1.45)
Form			
Whitening Pen	0.12 (0.48)	-1.21 (0.52)	0.11 (0.46)
Trays & Gel	0.40 (0.53)	-0.40 (0.55)	0.51 (0.49)
Light Technology	-1.39 (0.51)	-1.39 (0.51)	-0.79 (0.45)
Treatment Time	-0.36 (0.04)	-0.33 (0.04)	-0.37 (0.03)
Number of Treatments	-0.17 (0.03)	-0.18 (0.03)	-0.15 (0.02)
Time Until Results	-0.19 (0.03)	-0.21 (0.03)	-0.25 (0.03)
Peroxide %	0.21 (0.08)	0.21 (0.07)	0.06 (0.06)
Price	-0.44 (0.03)	-0.35 (0.03)	-0.32 (0.02)
ρ	-5.06 (0.40)	-4.90 (0.46)	-5.01 (0.39)

We study the joint effect of the probability of updating and the amount of learning about the value of the outside good in the next section.

Finally, we find that the model fit statistics, price coefficient and brand intercepts are smallest for Condition 3 in the tooth-whitening datasets. Across conditions, we find that the relative magnitude of these coefficients is consistent with the presence of a scale effect (Fiebig et al., 2010) that has previously been documented in the literature. We do not offer a specific reason for the finding that an increase in marketplace information is associated with an increase in the error scale of the choice model. We leave this as an issue for future research.

5.3. Inferences about the no-choice option

We investigate the increase in the selection of the outside good in our data using a model that allows the value of the outside good to increase across the choice tasks. We find evidence that the updating parameter improves the fit of the standard model and that updating the value of the outside good is sufficient for modeling consumer learning in this setting given the results shown in Fig. 5. We find that learning is isolated to the value of the outside good, which affects the estimated brand part-worths in a tradition conjoint choice model and not the part-worths for other product features.

Fig. 6 displays boxplots of the outside good utility for each respondent across choice tasks for both the MP3 and toothwhitening studies. The figure indicates that respondents in the MP3 player study are increasingly updating their outside good utility through the tasks, reflecting that they are learning about the market and the reference price when encountering new products. In contrast, respondents in the tooth-whitening study show only a small increase in outside good utility, indicating that updating occurs less often. The least amount of updating occurs in Condition 3 in which the respondents obtained the largest amount of market pricing and product information.

6. Discussion

The effect of the no-choice option has been discussed extensively in the behavioral decision theory (BDT) literature, but has not been thoroughly investigated in the context of conjoint analysis where respondents are confronted with multiple choice tasks from which part-worths are estimated. Conjoint analysis differs from models used in BDT research where attraction effects (Dhar, 1997) and choice uncertainty (Dhar and Simonson, 2003) have been shown to affect preferences for the no-choice option. In a conjoint analysis, attributes and their levels are defined in a concise and concrete manner (Orme, 2010), seeking to avoid the presence of task uncertainty. The instructions for expressing preferences in conjoint tasks ask respondents to imagine that unmentioned attributes are the same for each choice option, and therefore do not factor into the relative preferences for the choice options (see Gunasti and Ross (2008)). In addition, inferences about part-worths are made based on a dozen or more choice tasks, so that any contextual effect that may influence choice in one task is different than those in another choice task and are treated as part of the error term of the model. Therefore, a respondent wishing to avoid being too extreme in their expressed preferences, for example, will not have this tendency expressed in any specific part-worth (Simonson and Tversky, 1992). Similarly, attraction and dominance effects (Busemeyer and Townsend, 1993; Huber et al., 2014) will not be reflected in a particular part-worth because of multiple choice tasks. Conjoint studies also screen out respondents who are not familiar with the offerings in a product category, whereas BDT studies often do not employ screens. Respondents who pass the screening questions are assumed to be already familiar with the product features and the price of products charged in the marketplace, and can recall this information without being affected by the context of the choice task.

We find in our analysis that preference for the no-choice option is dynamically updated in the MP3 dataset where all offerings are new to the market and respondents lack a full understanding of the product and its benefits. Dynamic updating is still present

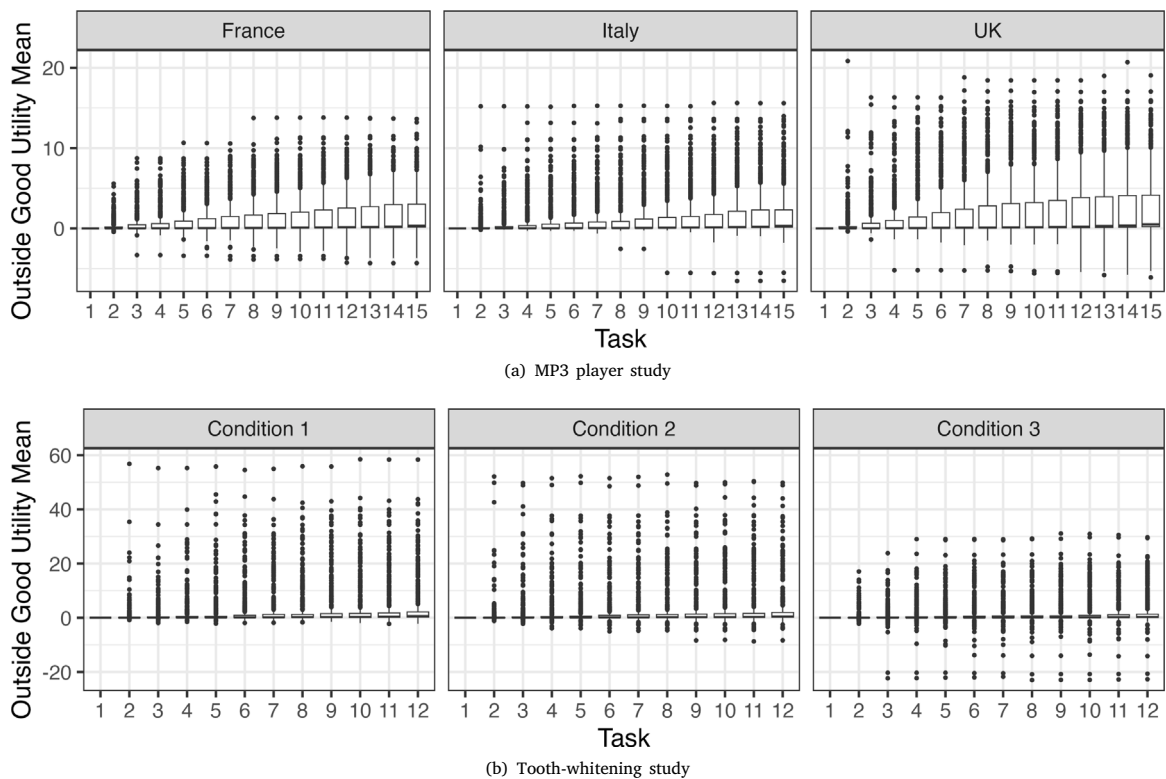


Fig. 6. The individual outside good utility means for each respondent in (a) MP3 player study and (b) tooth-whitening study.

in the teeth-whitening dataset but is significantly reduced and only present in the first two treatment conditions. This is consistent with a mature category wherein standard screening yields respondents with greater knowledge of prices and product offerings. In the third treatment condition these market details are clearly communicated before the conjoint study, so learning no longer occurs through the tasks and dynamic updating is eliminated.

We find from our analysis of the MP3 data, where respondents are less familiar with market prices and offerings, that preference for the no-choice option is dependent on the prices and features of the inside good options. In contrast to updating based on learning the trade-off between quality and price, as described in [Simonson and Tversky \(1992\)](#), we argue that respondents understand the relative differences between attribute levels (e.g., different display sizes), but not the no-choice option. In conjoint analysis, the brand part-worths are measured relative to outside good, which is set equal to zero in our analysis. This finding, that the part-worths for attribute-level are unaffected while the brand part-worths are affected by uncertainty about the market, is reflected in the model-free evidence shown in [Fig. 5](#) and in the updates to $\psi_{out,t}$. More importantly, we find that the price coefficient estimate is unaffected, indicating that respondents are using the price locally to compare alternative inside goods, but not globally. That is, they allow price to impact their choice within a subset of available options (secondary demand) but do not consider the broader implications of price on overall market affordability (primary demand).

An implication of our findings relates to the design of choice-based conjoint tasks. Our results suggest that researchers and practitioners should consider the potential impact of learning when including the no-choice option, even when the inside good options are hypothetical. We argue that even for familiar products like tooth-whitening products, using appropriate video warm-ups and providing additional market information can help mitigate the updating phenomenon across tasks.

7. Conclusion

This paper reexamines the no-choice option in conjoint analysis and finds that preference for it is affected by the respondent's familiarity with the product category. When the general level of prices is not known, as in the MP3 dataset, respondents update their preference for the outside good as they learn about the distribution of market prices. In the teeth-whitening dataset, dynamic updating is significantly reduced, in part due to consumer familiarity with the category. All dynamic updating is eliminated when information about products and prices in the market is shared prior to the study. We present evidence that non-brand part-worth estimates are unaffected, while the brand part-worths are affected in a uniform manner.

Our results reinforce the idea that a properly designed and executed conjoint study, where respondents are screened for inclusion based on product familiarity, product features are described in concrete and unambiguous terms, respondents are instructed to

assume missing attributes are the same for all choice alternatives, and a training video is presented to respondents so that they are familiar with the choice task, is largely immune to potential biases identified in laboratory experiments that lack these qualities. The no-choice option is needed in conjoint analysis because of the limited number of choice alternatives that can be presented to respondents in a choice task and carries with it the implicit assumption that respondents are aware of the prices charged for choice alternatives in the marketplace. Informing respondents of these prices prior to the conjoint choice task is encouraged, but may not be necessary if prices are generally known.

Our results are consistent with the notion of dynamic learning about the value of the product offerings in a competitive market, but they are not conclusive about its nature. The reason is because we do not have process evidence that would allow us to build a more formal model of what the outside good represents. We present evidence that updating occurs in a probabilistic manner, but do not provide evidence that relates it to specific variables or to give the outside good meaning beyond not making a selection of an inside good. More research is needed to isolate the source of learning to move beyond our simple model that assumes a constant updating probability for the outside good.

CRedit authorship contribution statement

Cheng-Yu Hung: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation. **Peter Kurz:** Writing – original draft, Supervision, Resources, Investigation, Data curation. **Roger A. Bailey:** Writing – original draft, Supervision, Resources, Investigation, Data curation. **Joel Huber:** Writing – original draft, Supervision, Project administration, Conceptualization. **Greg M. Allenby:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Screening questions used in tooth whitening study

To be included in the survey, respondents were required to make their own household hygiene product purchases, be medically eligible to use the product, and be currently engaged in or actively considering a purchase in the relevant product category. The screening questions used to assess these criteria are presented below.

Which of the following describes your level of involvement in purchasing hygienic products (soap, shampoo, toothpaste, etc.) in your household? ☐ I do all of the purchasing of hygienic products in my household ☐ I share the responsibility of purchasing hygienic products in my household ☐ I am not involved in purchasing hygienic products in my household	Please select all of the following dental/orthodontic work you CURRENTLY have or PLAN to have done in the next three months. ☐ Braces ☐ Dentures ☐ Caps, crowns, veneers, or fillings - not visible when smiling (back teeth) ☐ Caps, crowns, veneers, or fillings - visible when smiling (front teeth)
Have you purchased a teeth whitening product in the past? ☐ Yes ☐ No	Would you consider buying a teeth whitening product? ☐ Yes ☐ No

Appendix B. Binomial logit model detail estimation in preliminary analysis

For iteration d of the Monte Carlo draws, let $(\beta_h^{(d)}, \psi_{zh}^{(d)})$ denote the posterior draw of the part-worth vector and the marginal utility of money for respondent h , respectively. The inclusive value for the inside goods in task t for respondent h is computed as

$$W_{ht}^{(d)} = \ln \sum_{k=1}^J \exp \left(\beta_h^{(d)'} a_{hkt} - \psi_{zh}^{(d)} p_{kt} \right),$$

where a_{hkt} denotes the vector of attribute levels and p_{kt} is the price of the k th inside good out of a total of J inside goods in choice task t . Let Y_{ht} be the dependent variable, where $Y_{ht} = 1$ if respondent h selects the outside good in task t , and $Y_{ht} = 0$ otherwise. We fit a binomial logit model with $Y_{ht} \sim \text{Bernoulli} \left(p_{ht}^{\text{out},(d)} \right)$, where

$$\text{logit} \left(p_{ht}^{\text{out},(d)} \right) = \alpha_0^{(d)} + \alpha_1^{(d)} W_{ht}^{(d)} + \alpha_2^{(d)} (\text{task number}),$$

where $p_{ht}^{\text{out},(d)}$ is the probability of selecting the outside good for respondent h in task t , and the task number takes integer values from 1 to 15, corresponding to the number of choice tasks. We report the mean and standard deviation of $(\alpha_0^{(d)}, \alpha_1^{(d)}, \alpha_2^{(d)})'$ across the last 1,000 iterations of the Monte Carlo draws in Table 4.

Appendix C. Generating process

1. Draw $\phi_h = (\beta_h, \psi_{zh}, \rho_h) \sim \text{Normal}(\bar{\phi}, \Sigma_\phi)$ for respondent h .

β_h is the vector of preference of each attribute level for respondent h , ψ_{zh} is the marginal utility for money, and ρ_h is the parameter for updating the outside good utility in each task. In each task t , respondent h receives J inside good options with x_{hjt} the attribute vector and p_{hjt} the price of the inside good j . In the following, h will be omitted for simplicity.

2. For task 1,

- Deterministic term of the inside good j : $V_{j1} = \beta' x_{j1} - \psi_z p_{j1}$
- Deterministic term of the Outside good: $V_{out,1} = 0$ for identification issue.
- Choose inside good j with probability

$$P(y_1 = j) = \frac{\exp(V_{j1})}{1 + \sum_{l=1}^J \exp(V_{l1})}$$

and choose the outside good with probability

$$P(y_1 = out) = \frac{1}{1 + \sum_{l=1}^J \exp(V_{l1})}$$

3. For task $t = 2, \dots, T$,

- Deterministic term of the inside good j : $V_{jt} = \beta' x_{jt} - \psi_z p_{jt}$
- Update outside good utility:
 - (a) Draw a latent state s_{t-1} for indicating updating the outside good utility or not:

$$s_{t-1} = \begin{cases} 1, & \text{with probability } \frac{\exp(\rho)}{1 + \exp(\rho)} \\ 0, & \text{o.w.} \end{cases}$$

- (b) Calculate the deterministic term of the outside good utility as:

$$V_{out,t} = \begin{cases} \ln[\sum_{j=1}^J \exp(V_{j,t-1})], & s_{t-1} = 1 \\ V_{out,t-1}, & s_{t-1} = 0 \end{cases}$$

- Choose inside good j with probability

$$P(y_t = j) = \frac{\exp(V_{jt})}{\exp(V_{out,t}) + \sum_{l=1}^J \exp(V_{lt})}$$

and choose the outside good with probability

$$P(y_t = out) = \frac{\exp(V_{out,t})}{\exp(V_{out,t}) + \sum_{l=1}^J \exp(V_{lt})}$$

4. Repeat Steps 1–3 for each respondent h .

Appendix D. Estimation algorithm

The priors are represented as

$$\begin{aligned} (\beta_h, \psi_{zh}, \rho_h)^T &\sim N(\bar{\phi}, \Sigma_\phi) \\ \bar{\phi} &\sim N(0, 100I) \\ \Sigma_\phi &\sim IW(v, \Delta) \end{aligned}$$

where $v = L + 1 + 3$, $\Delta = vI$, and L denotes the number of possible attribute levels excluding price.

1. Generate $\beta_h, \psi_{zh}, \rho_h | \bar{\phi}, \Sigma_\phi, y_h, X_h, \{s_{ht}\}$ for $h = 1, \dots, N$:

We use a random-walk chain M-H step. Let $\phi_h^{(n)} = (\beta_h^{(n)}, \psi_{zh}^{(n)}, \rho_h^{(n)})^T$ represent a new candidate vector, and let $\phi_h^{(o)} = (\beta_h^{(o)}, \psi_{zh}^{(o)}, \rho_h^{(o)})^T$ represent the old vector from the previous iteration of the chain.

Generate $\phi_h^{(n)} \sim N(\phi_h^{(o)}, s^2 \Sigma_\phi + I)$ where $s = 2.93/\sqrt{L+1}$ (see Rossi et al. (2024)), and accept the new candidate with the probability,

$$AR = \min \left[\frac{L_h(\phi_h^{(n)})}{L_h(\phi_h^{(o)})}, 1 \right]$$

where $L_h(\phi_h) = P(y_h | X_h, \beta_h, \{s_{ht}\}) \{ \prod_{t=1}^{T-1} P(s_{ht} | \rho_h) \} P(\phi_h | \bar{\phi}, \Sigma_\phi)$

2. Generate $\{s_{ht}\}_{t=1}^{T-1} | \bar{\phi}, \Sigma_{\phi}, y_h, X_h, \beta_h, \psi_{zh}, \rho_h$ for $h = 1, \dots, N$:
Generate $s_{ht}^{(n)} \sim \text{Bernoulli}(p_h)$ for $t = 1, \dots, T-1$ and accept the new candidate with the probability,

$$AR = \min \left[\frac{L'_h(\{s_{ht}\}^{(n)})P(\{s_{ht}\}^{(o)}|p_h)}{L'_h(\{s_{ht}\}^{(o)})P(\{s_{ht}\}^{(n)}|p_h)}, 1 \right]$$

where $p_h = 0.05$, $L'_h(\{s_{ht}\}) = P(y_h | X_h, \beta_h, \{s_{ht}\}) \{ \prod_{t=1}^{T-1} P(s_{ht} | \rho_h) \}$, $P(\{s_{ht}\} | p_h) = \prod_{t=1}^{T-1} p_h^{s_{ht}} (1 - p_h)^{1-s_{ht}}$. We choose $p_h = 0.05$, as it produces an acceptance rate of approximately 0.4; we note that $p_h = 0.01$ also performs adequately.

3. Generate $\bar{\phi} | \phi_h, \Sigma_{\phi}$ and $\Sigma_{\phi} | \phi_h, \bar{\phi}$:
Similar algorithm to the Bayesian linear regression Metropolis–Hastings algorithm.

Data availability

Data will be made available on request.

References

- Bahamonde-Birke, Francisco J., Navarro, Isidora, de Dios Ortúzar, Juan, 2017. If you choose not to decide, you still have made a choice. *J. Choice Model.* 22, 13–23.
- Balbontin, Camila, Hensher, David A., Collins, Andrew T., 2017. Do familiarity and awareness influence voting intention: The case of road pricing reform? *J. Choice Model.* 25, 11–27.
- Becker, Karolin, Zweifel, Peter, 2008. Age and choice in health insurance: evidence from a discrete choice experiment. *Patient* 1 (1), 27–40.
- Ben-Akiva, Moshe, McFadden, Daniel, Train, Kenneth, 2019. Foundations of stated preference elicitation: Consumer behavior and choice-based conjoint analysis. *Found. Trends® Econ.* 10 (1–2), 1–144.
- Boxall, Peter, Adamowicz, Wiktor L., Moon, Amanda, 2009. Complexity in choice experiments: choice of the status quo alternative and implications for welfare measurement. *Aust. J. Agric. Resour. Econ.* 53 (4), 503–519.
- Brazell, Jeff D., Diener, Christopher G., Karniouchina, Ekaterina, Moore, William L., Séverin, Valérie, Uldry, Pierre-Francois, 2006. The no-choice option and dual response choice designs. *Mark. Lett.* 17 (4), 255–268.
- Busemeyer, Jerome R., Townsend, James T., 1993. Decision field theory: a dynamic-cognitive approach to decision making in an uncertain environment. *Psychol Rev* 100 (3), 432.
- Carson, Richard T., Louviere, Jordan J., Anderson, Donald A., Arabie, Phipps, Bunch, David S., Hensher, David A., Johnson, Richard M., Kuhfeld, Warren F., Steinberg, Dan, Swait, Joffe, Timmermans, Harry, Wiley, James B., 1994. Experimental analysis of choice. *Mark. Lett.* 5 (4), 351–367.
- Dhar, Ravi, 1997. Consumer preference for a no-choice option. *J. Consum. Res.* 24 (2), 215–231.
- Dhar, Ravi, Simonson, Itamar, 2003. The effect of forced choice on choice. *J. Mark. Res.* 40 (2), 146–160.
- Enke, Benjamin, Graeber, Thomas, 2023. Cognitive uncertainty. *Q. J. Econ.* 138 (4), 2021–2067.
- Fiebig, Denzil G., Keane, Michael P., Louviere, Jordan, Wasi, Nada, 2010. The generalized multinomial logit model: Accounting for scale and coefficient heterogeneity. *Mark. Sci.* 29 (3), 393–421.
- Green, Paul E., Rao, Vithala R., 1971. Conjoint measurement for quantifying judgmental data. *J. Mark. Res.* 35, 5–363.
- Gunasti, Kunter, Ross, William T., 2008. How inferences about missing attributes decrease the tendency to defer choice and increase purchase probability. *J. Consum. Res.* 35 (5), 823–837.
- Haaajjer, Rinus, Kamakura, Wagner, Wedel, Michel, 2001. The 'no-choice' alternative to conjoint choice experiments. *Int. J. Mark. Res.* 43, 93–106.
- He, Junnan, 2024. Bayesian contextual choices under imperfect perception of attributes. *Manag. Sci.* 70 (3), 1465–1482.
- Hess, Stephane, Hensher, David A., Daly, Andrew, 2012. Not bored yet – revisiting respondent fatigue in stated choice experiments. *Transp. Res. Part A: Policy Pr.* 46 (3), 626–644.
- Huber, Joel, Payne, John W., Puto, Christopher P., 2014. Let's be honest about the attraction effect. *J. Mark. Res.* 51 (4), 520–525.
- Hung, Cheng-Yu, Liu, YiChun Miriam, Brazell, Jeff D., Allenby, Greg M., 2025. Dual response in conjoint analysis. *Mark. Lett.*
- Khaw, Mel Win, Li, Ziang, Woodford, Michael, 2020. Cognitive imprecision and small-stakes risk aversion. *Rev. Econ. Stud.* 88 (4), 1979–2013.
- Kim, Youngju, Hardt, Nino, Kim, Jahwan, Allenby, Greg M., 2022. Conjoint screening in models of multiple discreteness. *Int. J. Res. Mark.* 39 (4), 1209–1234.
- Krueger, Rico, Rashidi, Taha H., Rose, John M., 2016. Preferences for shared autonomous vehicles. *Transp. Res. Part C: Emerg. Technol.* 69, 343–355.
- Kuhfeld, Warren F., 2010. *Marketing Research Methods in SAS*. SAS Institute Incorporated., URL https://support.sas.com/resources/papers/tnote/tnote_marketresearch.html.
- Louviere, Jordan J., Flynn, Terry N., Carson, Richard T., 2010. Discrete choice experiments are not conjoint analysis. *J. Choice Model.* 3 (3), 57–72.
- Manski, Charles F., McFadden, Daniel, et al., 1981. *Structural Analysis of Discrete Data with Econometric Applications*. MIT Press Cambridge, MA.
- Natenzon, Paulo, 2019. Random choice and learning. *J. Politi. Econ.* 127 (1), 419–457.
- Newton, Michael A., Raftery, Adrian E., 1994. Approximate bayesian inference with the weighted likelihood bootstrap. *J. R. Stat. Soc. Ser. B Stat. Methodol.* 56 (1), 3–26.
- Orme, Bryan K., 2010. *Getting Started with Conjoint Analysis: Strategies for Product Design and Pricing Research*. Research Publishers LLC, Madison.
- Orme, Bryan, Chrzan, Keith, Allenby, Greg, 2023. Managerial and academic considerations for three approaches to willingness to pay. In: *2023 Sawtooth Software Proceedings*. pp. 121–130.
- Pachali, Max J., Kurz, Peter, Otter, Thomas, 2017. The perils of ignoring the budget constraint in single-unit demand models. *Tech. rep. SSRN Working Paper*.
- Parker, Jeffrey R., Schrieff, Rom Y., 2011. Rejectable choice sets: How seemingly irrelevant no-choice options affect consumer decision processes. *J. Mark. Res.* 48 (5), 840–854.
- Rao, Vithala R., 2014. *Applied Conjoint Analysis*. Springer.
- Rose, John M., Bliemer, Michiel C.J., Hensher, David A., Collins, Andrew T., 2008. Designing efficient stated choice experiments in the presence of reference alternatives. *Transp. Res. Part B: Methodol.* 42 (4), 395–406.
- Rossi, Peter E., Allenby, Greg M., Misra, Sanjog, 2024. *Bayesian Statistics and Marketing*. John Wiley & Sons.
- Salkeld, G., Ryan, M., Short, L., 2000. The veil of experience: do consumers prefer what they know best? *Heal. Econ.* 9 (3), 267–270.
- Sandorf, Erlend Dancke, Campbell, Danny, Hanley, Nick, 2017. Disentangling the influence of knowledge on attribute non-attendance. *J. Choice Model.* 24, 36–50.
- Scarpa, Riccardo, Ferrini, Silvia, Willis, Kenneth, 2005. Performance of Error Component Models for Status-Quo Effects in Choice Experiments. *Springer Netherlands, Dordrecht*, pp. 247–273.

- Simonson, Itamar, Tversky, Amos, 1992. Choice in context: Tradeoff contrast and extremeness aversion. *J. Mark. Res.* 29 (3), 281–295.
- Train, Kenneth E., 2009. *Discrete Choice Methods with Simulation*, second ed. Cambridge University Press.
- Train, Kenneth, Wilson, Wesley W., 2008. Estimation on stated-preference experiments constructed from revealed-preference choices. *Transp. Res. Part B: Methodol.* 42 (3), 191–203.
- Tversky, Amos, Shafir, Eldar, 1992. Choice under conflict: The dynamics of deferred decision. *Psychol. Sci.* 3 (6), 358–361.
- Van Loo, Ellen J., Caputo, Vincenzina, Nayga, Rodolfo M., Seo, Han-Seok, Zhang, Baoyue, Verbeke, Wim, 2015. Sustainability labels on coffee: consumer preferences, willingness-to-pay and visual attention to attributes. *Ecol. Econ.* 118, 215–225.